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Critical heat flux for SiC- and Cr-coated plates under atmospheric condition

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ABSTRACT

Hydrogen gas released by Zircaloy's oxidative reaction can lead to disastrous results, as observed in the Fukushima accident. SiC and Cr, however, have low potential for a hydrogen explosion. This paper is focused on measuring the critical heat flux (CHF) for SiC and Cr surfaces, which are known to have favorable properties such as oxidation resistance of a bulk material in atmospheres containing steam or a high vapor fraction. SiC also has a low neutron absorption cross-section, which is one of the required characteristics for cladding of light water reactors (LWRs).

Regarding their application, SiC and Cr materials can potentially be used to design coatings for the Zircaloy cladding that is used at present. We suggest that SiC coating can be achieved through a PVD (physical vapor deposition)-sputtering and Cr coating can be achieved through an electroplating process on the surface of the Zircaloy cladding. After coating, reaction between the Zircaloy cladding and steam will be reduced. In this study, coatings were achieved on the stainless steel plate to assess the surface effects on CHF. Two groups of experiments were conducted to assess the heat transfer performance of the SiC and Cr surfaces. The experiments were carried out under pool boiling condition of saturated de-ionized (DI) water to assess the SiC- and Cr-coated surfaces. SiC and Cr coatings with two levels of thickness were prepared and evaluated. The SiC-coated surfaces showed improved CHF compared with the other surfaces used in our experiments. Test sections with thicker coatings showed higher CHF results than the thinner ones. The enhanced performance was due to improved surface wettability. On the other hand, Cr-coated surfaces have shown lower CHF results compared with other surfaces regardless of the thickness used in our experiment.

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1. Introduction

During severe accidents or LOCA (loss-of-coolant accident) situation when coolant is not properly provided, claddings are often exposed to steam. Under such condition, the integrity of the cladding can be threatened by accelerated embrittlement through hydriding and active oxidation. If hydrogen is not properly eliminated, explosions can occur which may threaten the integrity of the containment. The Fukushima accident in Japan demonstrates the potential danger of actively produced hydrogen not properly eliminated and exothermic reaction between the zirconium-based alloy and steam. Fukushima accident provoked interest in materials and structures that minimize oxidation, even at a high temperature. As a candidate, SiC has been regarded as a strong candidate material because it undergoes less oxidation than the Zircaloy claddings that are used at present. SiC also has a low neutron absorption cross-section, which is a necessary property for LWR claddings. Also, Cr is a well-known corrosion-resistant material at high temperatures and commonly used in industries. Corrosion-resistant materials such as SiC and Cr reduce oxidation by way of forming protective oxide layer on the surface which inhibits further oxidation. Historically, SiC has been used under high temperature conditions such as in fusion reactors and other very high temperature reactors due to its excellent properties at high temperatures. In addition, SiC monoliths and SiC/SiC composites have been investigated for use in LWRs (Carpenter, 2006; Feinroth et al., 2009; Jung et al., 2012; Kim et al., 2012). According to Feinroth et al. (2009), triplex specimens have the same hoop strength as unirradiated samples within 2 standard deviations after exposure to typical PWR operating conditions, including fast neutron irradiation. Furthermore, their study showed that the addition of an outer composite layer with a unique fiber architecture and an outer

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environmental barrier layer can provide additional pressure retention capacity. Kim et al. (2012) discussed overall trends and future studies regarding the applicability of SiC for LWR fuel. Jung et al. (2012) introduced metal-ceramic hybrid cladding consisting of an inner zirconium tube and an outer SiC fiber-matrix made from SiC ceramic composite. This study attempted to resolve the problems associated with fission gas confinement and the joining of end caps. In the study by Carpenter (2006), SiC duplex was proposed as a fuel cladding. In this study, the behavior of SiC cladding was compared to that of conventional Zircaloy cladding to demonstrate that SiC has superior resistance to creep and mechanical degradation due to radiation or oxidation. Cr has also been widely used as a corrosion-resistant material since it forms protective oxide layer on the surface. Not just under the air and water (steam) conditions, chromium oxide is also stable in alkaline solutions. Many studies have confirmed the effect of Cr-added alloys and the amount of Cr addition to the bulk materials. In many industrial areas, Cr is included in alloy or is used as a coating material to protect the bulk material from corrosion and wear. According to Viswanathan et al. (2006), up to 40% Cr is recommended for highly corrosive environments. According to Rohr (2005), one possible way to minimize the corrosion phenomena is Cr coating on the surface, and the bulk material will give mechanical properties like creep resistance at the same time. Kim et al. (2013) have carried out stainless steel oxidation experiments with different Cr contents under 900 °C steam condition up to 200 h. The results have shown that specimen of high Cr content formed protective layer and prevented further oxidation. They suggested minimum Cr content of 18–20 wt% is required to act as an oxidation barrier.

Under nucleate boiling conditions, heat can be transferred into liquids efficiently with less superheating on the surface. However, as heat flux on the surface increases further, bubbles rapidly form and begin to merge with each other, inhibiting the motion of liquid near the wall. Slugs and vapor become more unstable with increasing surface temperature and begin to form local blankets. The proportion of film increases and the heat transfer coefficient (HTC) decreases with further increases in the surface temperature. At a certain point referred to as the CHF, the surface is completely covered by a vapor film. At this point, the temperature of the surface rises abruptly due to a sudden decrease in the HTC caused by the film layer. Above the CHF, an abrupt temperature increase can induce material failure. Therefore, the CHF is a very important value related to thermal hydraulic phenomena.

CHF is related to many factors including material properties and thickness. Many studies have shown that surface wettability is one of the most important factors that influence the CHF. According to Phan et al. (2009), wettability is an important factor in bubble growth. Kandlikar (2001) proposed a theoretical CHF model that used the dynamic contact angle to assess the effect of surface wettability. According to the experimental results of Jeong et al. (2008), the CHF is strongly affected by wettability, as determined by contact angle measurements. In the studies by Lee et al. (2012, 2013) magnetite nanoparticles deposited on a heated surface improved both wettability and CHF. According to other previous studies (Chun et al., 2011; Kim et al., 2006; Lee et al., 2012; Song et al., 2014), CHF was enhanced in nano-fluids due to the deposition of nano-particles on the surface during evaporation. Many other studies have verified this relationship between wettability and CHF, making it a major subject of study in the heat transfer characteristics of materials. In addition to wetting, thickness of the material also greatly affects heat transfer. According to Gogonin (2009), CHF depends substantially on the thickness of heated wall. This study has summarized based on many studies that heat generated on a thick-walled heater is dissipated both to coolant and heater itself by conduction. Tachibana et al. (1967) also showed that there exists great dependency of CHF on wall thickness introducing the newly defined parameter. But this and other studies showed that there is some saturated thickness from which thickness effect is negligible. Based on these studies, it is clear that changing the surface of claddings from Zircaloy to SiC or Cr will change the heat transfer properties. However, there is still a lack of information regarding the thermal hydraulic properties of SiC and Cr coatings. Therefore, the purpose of this study is to measure the CHF of SiC and Cr surfaces to assess their applicability as coatings for LWR claddings. CHF values of SiC- and Cr- coated surfaces were measured for two different coating thicknesses under pool boiling conditions. Furthermore, the contact angle, scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS), atomic force microscopy (AFM), and X-ray diffraction (XRD) were used to evaluate heat transfer mechanisms and confirm the coating quality.

1.1. Preliminary calculation of neutron economy

Neutron economies were calculated for SiC- and Cr-coated Zircaloy cladding. For the purpose, MCNP calculation under assumption of infinite lattice was used. MCNP is abbreviation of Monte Carlo N-Particle code which is Monte Carlo transport code for several types of particles. Monte Carlo method can depict statistical reactions with given cross-section values from libraries like ENDF/B-VI and is useful for the complex calculations that is hard to be solved with deterministic methods. According to the calculation, SiC-coated Zircaloy cladding showed better result than that from the Cr-coated case. Chromium has relatively high neutron absorption cross section that chromium coatings on the Zircaloy surface will decrease the effective multiplication factor with increasing thicknesses more than the SiC coatings resulting in degraded neutron economy in LWR ($\sigma_{a,Cr}$ = 3.1b, $\sigma_{a,Si}$ = 0.16b, $\sigma_{a,C}$ = 0.0034b at 0.0253 eV). The schematic structure used in our calculation is shown in Fig. 1. The relative deviation of the multiplication factor when SiC and Cr coatings are added on zirconium-based alloy cladding are represented in Fig. 2. Relative deviation of the multiplication factor was calculated by relative error based on the multiplication factor of the non-coated zirconium-based cladding (k_{ref}).

2. Experimental apparatus

2.1. Test pool and procedure

The pool boiling experiment was carried out with DI water, and the fluid was saturated under atmospheric conditions. The condenser transformed the vapor into liquid and kept the system

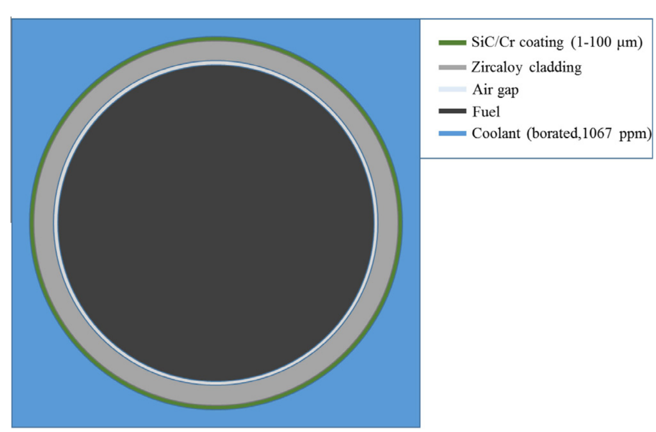


Fig. 1. The schematic description of the structures used in our calculation.

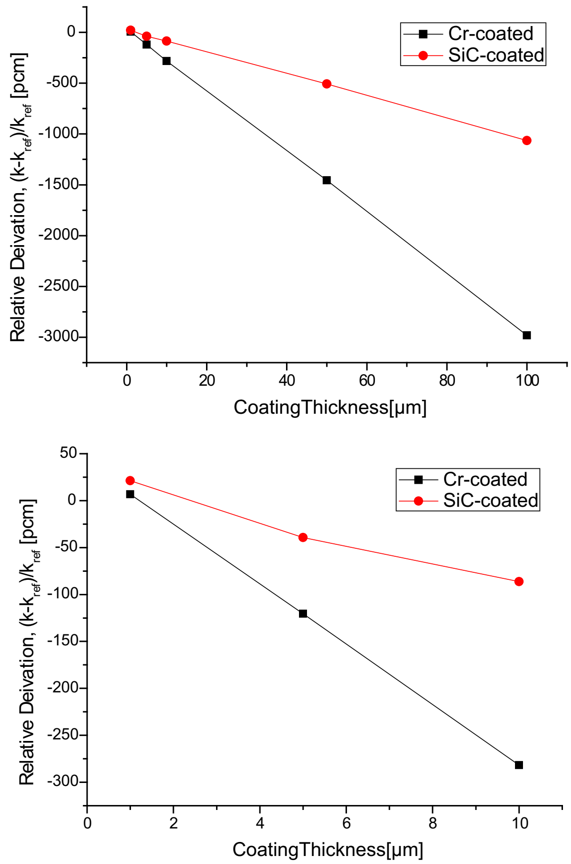


Fig. 2. The relative deviation of the multiplication factors with coating thickness.

under atmospheric conditions. Pre-heaters were situated below the test section to heat the fluid to maintain the saturated conditions. The surface temperature of the test section was measured through three K-type thermocouples (TC) attached to the downward surface of the substrate. The downward surface was insulated with silicone rubber and bakelite below it. The bulk temperature of the fluid was measured with an inserted K-type TC. The test section was heated directly using a DC rectifier with a maximum capacity of 45 kW (15 V, 3000 A). The test pool was made of polycarbonate. The test pool and the test section are shown in Fig. 3.

The experimental data including voltage, current and temperature were measured with a data acquisition system (DAS) consist of an Agilent 34,980 A data acquisition/control unit and a personal computer. Test sections were attached to copper electrodes, and they were cleaned by methyl alcohol before the experiments. Heat flux was gradually increased with electric power, and the resistance of the test section sharply increased with local burnout on the surfaces when CHF occurred. The CHF was calculated by dividing the amount of power supplied by the area of heat transfer once the CHF was reached. We regarded that the downward side of the test section was well insulated since no bubble could be seen generated from the bottom side of the heaters because the silicon rubber was situated on the downward surface, and the bakelite was located under it. For the DC heating method, CHF can be calculated using Joule's equation as follow:

$$q_{\rm CHF} = v_{\rm CHF} \times 1_{\rm CHF}/A_h$$

where, $q_{\rm CHF}$ is the CHF of the test section; $V_{\rm CHF}$ and $I_{\rm CHF}$ are the applied voltage and current, respectively, when the CHF is reached; and $A_{\rm h}$ is the heat transfer area.

Root-sum square (RSS) error is given by the individual uncertainties from the independent data. For the power meter used in the experiment, the uncertainty of the current and the voltage were 1.0% and 0.2%, respectively. The uncertainty from the heater

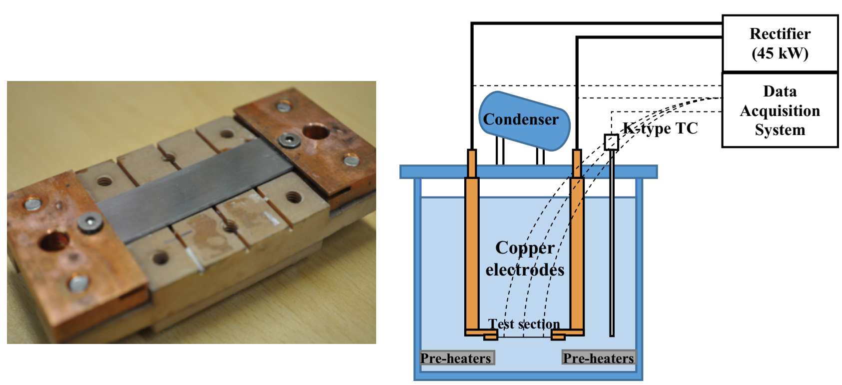
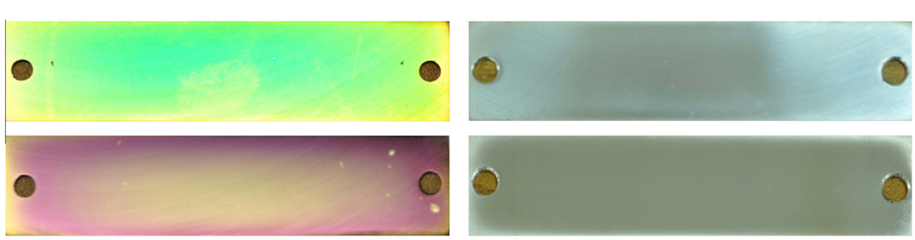


Fig. 3. Design of the heater and pool boiling apparatus.



(a) (b)

Fig. 4. Test sections used in the pool boiling experiments. (a) 400 nm, 1 μm thicknesses SiC surfaces and (b) 400-nm, 1 μm thicknesses Cr surfaces.

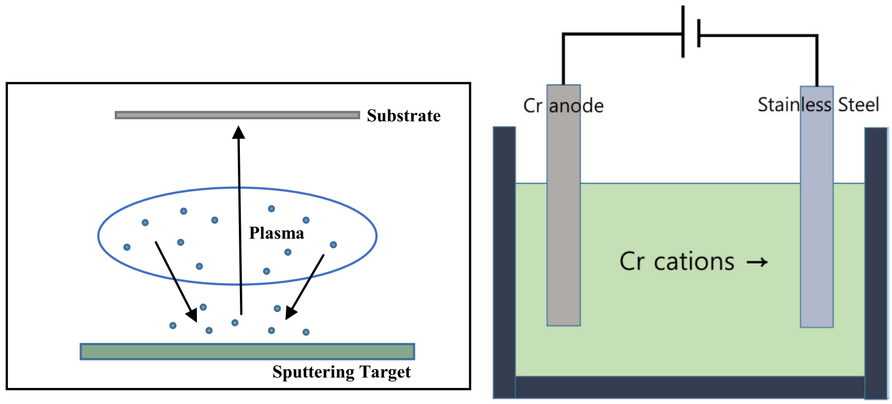


Fig. 5. The schematic of the PVD-sputtering and electroplating process.

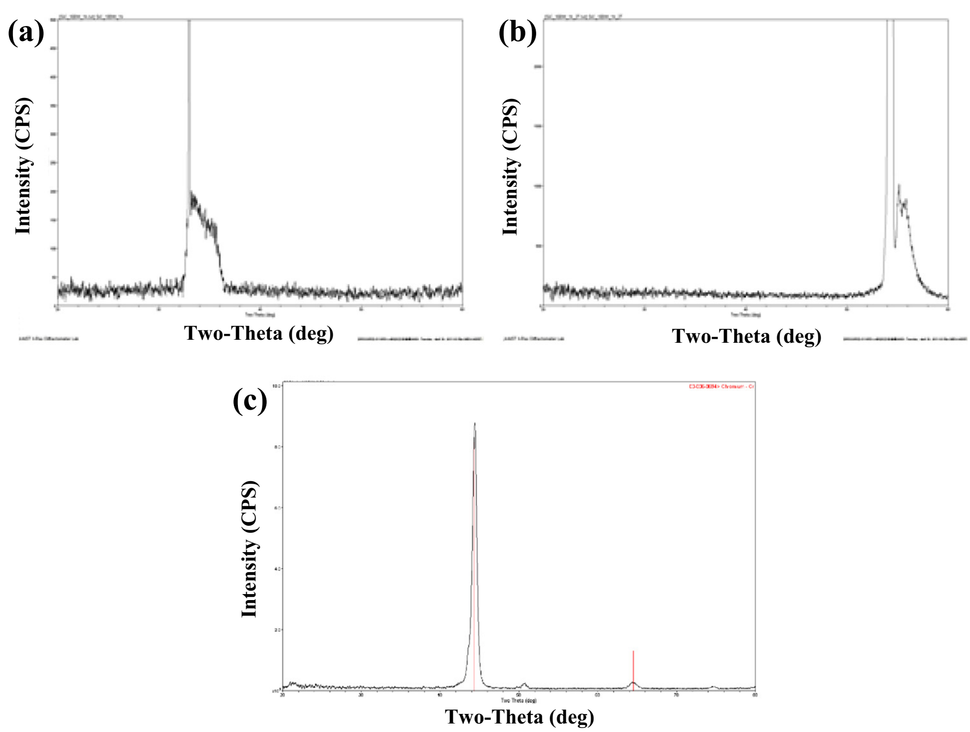


Fig. 6. XRD for the SiC-coated silicon wafer (amorphous) and Cr-coated stainless steel (cubic). (a) $\theta/2\theta$ scan (SiC), (b) 2θ scan (SiC) and (c) 2θ scan (Cr).

Table 1
EDS results for the SiC- and Cr-coated surfaces.

Element	C	O	Si
400 nm			
At%	32.67	21.07	46.26
Wt%	19.34	16.62	64.04
1 μm			
At%	32.01	23.9	44.09
Wt%	19.17	19.07	61.75

Element	O	Cr
1 μm		
At%	6.7	93.3
Wt%	2.16	97.84
10 μm		
At%	6.85	93.15
Wt%	2.21	97.79

area was 0.2%. Therefore, uncertainty of the heat flux was about 1.0%. Since the heat loss from the heater was approximately 2.3%, the overall uncertainty of the CHF was 3.3%.

2.2. Test section

Figures of the test sections are shown in Fig. 4, and the dimensions of the test sections were 0.05 m × 0.015 m × 0.0012 m (length × width × thickness). SiC coating was achieved through PVD-sputtering on stainless steel plates for 1 h (approximately 400 nm thick) and 3 h of deposition (approximately 1 μm thick) at a power of 100 W power. The stainless steel was polished with #600 and #1200 sandpapers. Cr coating was achieved by electroplating for about 5 min (1 μm thick) and 50 min (10 μm thick) at a current density of 20 ASD, respectively. Zircaloy-4 plate was prepared to measure CHF reference value, and the dimension was 0.05 m × 0.015 m × 0.00055 m (length × width × thickness).

The PVD-sputtering and electroplating process are shown in Fig. 5. The PVD-sputtering begins with the collision between emitted electrons and inert gases such as Ar. When the chamber is filled with inert gas and a voltage is applied, electrons emitted from the cathode collide with the inert gas atoms and ionize them. The inert gas ions in the plasma are then accelerated toward the cathode, where the target material is held. When the ions collide with the surface of the target material, neutral target atoms are emitted and deposited on the substrate as a coating. One of the advantages of the PVD process over chemical vapor deposition (CVD) is its relatively low working temperature because it is difficult to treat Zircaloy materials at high temperatures. Electroplating is for the metallic coating that is layered on conducting materials. Substrate to be coated is the cathode, and the target material to be used as a coating material is the anode in electrolyte with dissolved metal cations. Dissolved metal cations form coating on the surface of the substrate with supplied electrons, and the target material dissolves, accordingly. The process is shown in Fig. 5. One side of the stainless steel was plated with Cr.

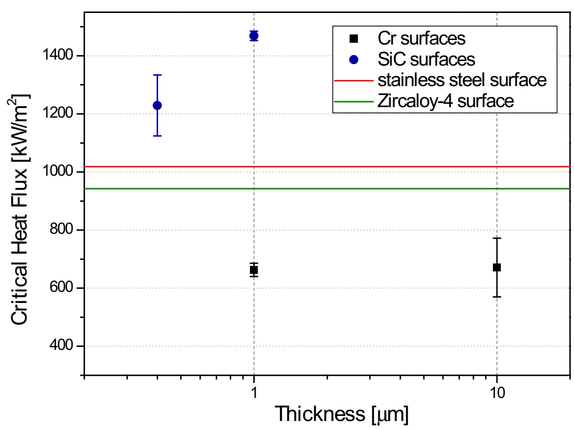


Fig. 7. CHF results for SiC- and Cr-coated plates, Zircaloy-4 plate and the stainless steel substrates.

To evaluate the composition and the state of the SiC and Cr coatings, EDS and XRD were used. EDS results showed the atomic or weight composition of the two kinds of coatings we have used (SiC and Cr). XRD was used to check the crystal structure of the coatings. For the XRD measurement, SiC coatings were achieved on the silicon wafer and Cr coatings were done on the bare stainless steel plate. According to the EDS and XRD results (Table 1, Fig. 6), the SiC coating was composed of Si/C/O in an amorphous

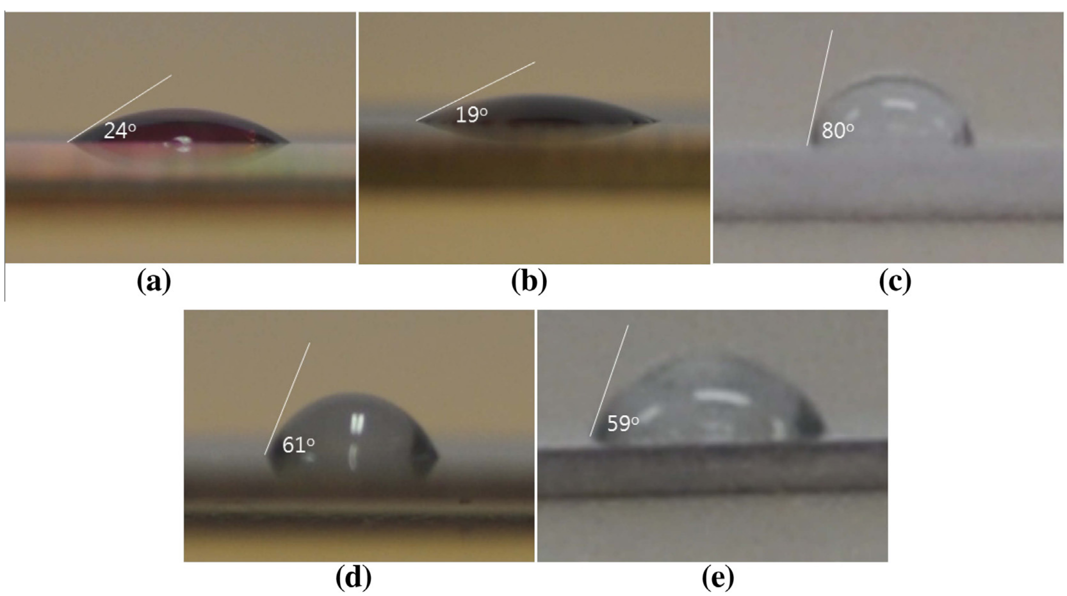


Fig. 8. Contact angles of (a) the 400 nm-thick SiC coating, (b) the 1 μm-thick SiC coating, (c) Cr-coated surface, (d) bare stainless steel and (e) Zircaloy-4 surface with 5 μL of water.

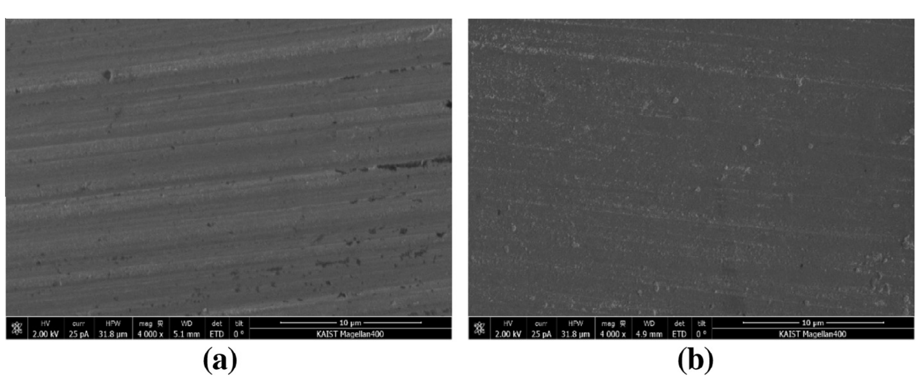
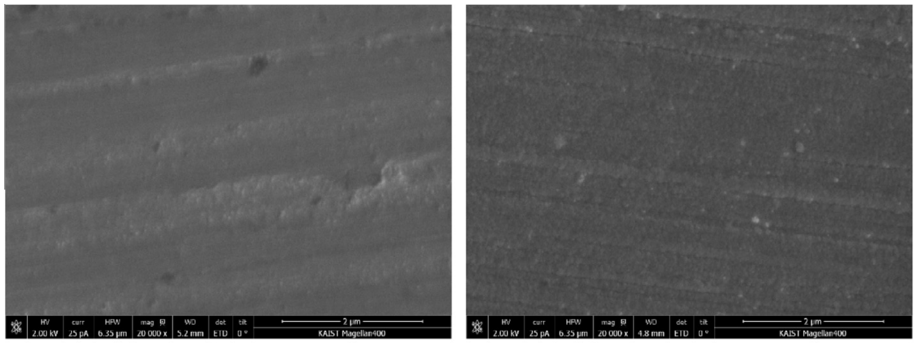
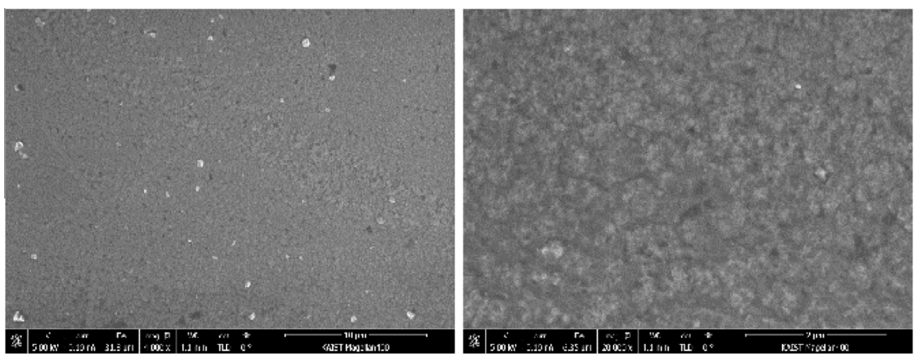


Fig. 9. SEM images (magnification 4000×) of (a) stainless steel and (b) the 400 nm thick SiC coating.



(a) (b)

Fig. 10. SEM images (magnification 20,000×) of (a) stainless steel and (b) the 400 nm thick SiC coating.



(a) (b)

Fig. 11. SEM images of 1 μm Cr-coated surfaces. Magnification of (a) 4000× and (b) 20,000×.

state, and the Cr coating was composed of Cr/O in cubic structure. In XRD result of SiC surface, plateau was arisen from the amorphous state, and the Si peak was due to the Si wafer. Peak in the XRD of Cr coating represents the cubic structure of the Cr coating.

3. Experimental results

SiC-coated surfaces have shown enhanced CHF results compared with bare stainless steel and Zircaloy-4 surfaces, regardless of the coating thickness. In contrast, Cr surfaces showed lower CHF values for both levels of thickness. CHF results are summarized in Fig. 7 with standard deviation bars. The experiments were done at least 3 times for each cases. The CHF of bare stainless steel was 1020 kW/m^2 with contact angle of 61° ± 5° before the experiments. The CHF of Zircaloy-4 was 940 kW/m^2 with contact angle of 59° ± 6°. There was a thickness effect for the SiC surfaces. 400 nm thickness SiC coating showed CHF value of 1230 kW/m^2 and 1 μm thickness for 1470 kW/m^2. For the Cr-coated surfaces, both 1 μm and 10 μm thickness showed similar values. Contact angles were measured before the experiments: 24° ± 6° for 400 nm thickness, 19° ± 4° for 1 μm thickness SiC coatings and 81° ± 4° for the Cr surfaces. Fig. 8

According to the AFM and SEM results (Fig. 9–11 and Table 2), there was no evident difference in roughness between the SiC- and Cr-coated surfaces and the bare stainless steel. However, due to the material characteristics of the coating surfaces, it displayed enhanced or reduced CHF.

According to Fig. 12, for the same thickness of coatings (1 μm), SiC surface showed the highest CHF value followed by stainless steel surface, Zircaloy-4 surface and Cr surface in order. The reason

Table 2
AFM Result (20 μm × 20 μm).

		Roughness [nm]	Average [nm]
SiC	400 nm	22.714	21.496
	400 nm	19.508	
	1 μm	22.267	
Cr	1 μm	28.141	24.593
	1 μm	27.071	
	10 μm	24.185	
	10 μm	18.975	
Bare stainless steel		22.765	25.438
		26.741	
		26.809	

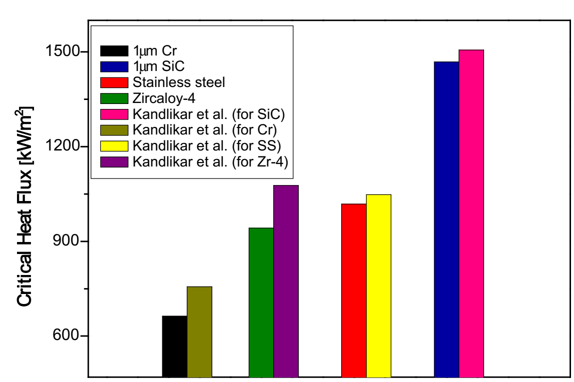


Fig. 12. CHF results for 1 μm thickness of SiC- and Cr-coated plates, Zircaloy-4 and the stainless steel substrate.

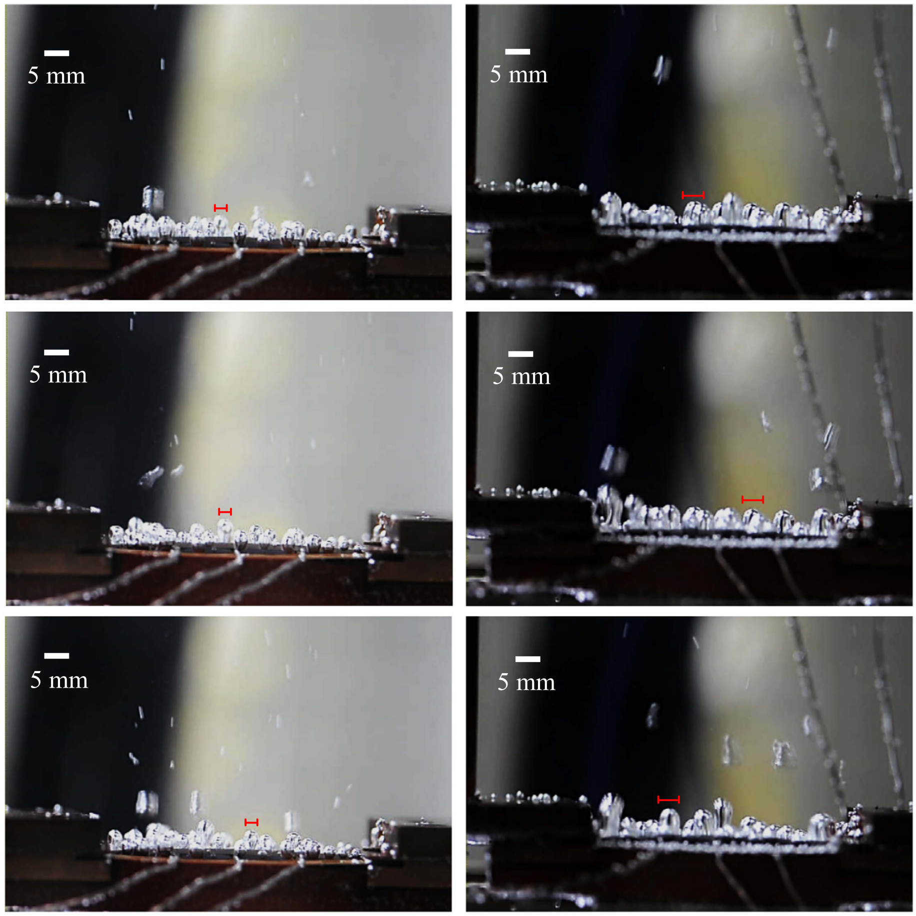


Fig. 13. Bubble generation pictures under the same heat flux conditions (left: stainless steel, right: Cr-coated surfaces).

for the results can be explained by the contact angle measurements. In addition to that, bubble generation pictures were taken for the stainless steel and Cr surfaces under 6 kW/m^2, 7 kW/m^2, 8 kW/m^2 heat flux conditions. According to Fig. 13, difference in bubble sizes between the Cr surfaces and the stainless steel surfaces were visually distinguishable, and the bubbles generated from the Cr surface are larger than that from the stainless steel surface (about 4 mm and 3 mm diameter of bubble sizes for Cr and stainless steel surfaces, respectively), which corresponds well with the contact angle measurement results. Larger bubbles have more chance to merge each other interrupting the water supply and higher chance of CHF occurrence accordingly. CHF values were compared with results acquired from Kandlikar (2001) using static contact angles measured before the experiments and the properties under atmospheric condition.

4. Conclusion and discussion

In this study, CHF was measured for SiC-coated, Cr-coated, bare stainless steel and Zircaloy-4 surfaces. SiC coatings and Cr coatings were achieved by PVD-sputtering and electroplating for two levels of thickness, respectively. SEM, EDS, AFM and XRD were used to evaluate the surface conditions, and contact angles were measured for each surface to assess the surface affinity with DI water with bubble generation pictures. According to the study, the SiC-coated surface showed enhanced CHF compared with other surfaces, with thicker SiC coatings showing better results. This enhancement was due to the hydrophilic character of the surface. Coletti et al. (2007) mentioned that SiC displays strong adhesion with water. In addition, several studies (Chang and Baek, 2003; Dhir and Liaw, 1989; Kim, 2012) have shown that surface oxidation enhances both the wettability and CHF. Therefore, the affinity of the SiC surface for water may correlate with the property of SiC itself and some amount of oxide present in the form of SiO_x. Cr-coated surface has shown lower CHF compared with other surfaces, and there was no thickness effect in our experimental range. The hydrophobic property could be shown by the contact angle measurement and the bubble generation pictures.

Among post-cladding candidates, both SiC and Cr surfaces are corrosion-resistant compared with conventional Zircaloy claddings. In addition, SiC surfaces have enhanced CHF compared with other surfaces in our experiments. Even though the surface roughness became similar to that of the bare stainless steel plate, the SiC-coated surface remained more hydrophilic, resulting in enhanced CHF. Therefore, SiC-coated surfaces transfer heat at least as well as or, in some cases, better than the materials that are presently used in LWRs. In case of the Cr surface, some special treatments are needed on the surface to reach the similar CHF range.

To minimize the oxidation of the cladding, it may be possible in the near future to coat SiC or Cr on conventional Zircaloy cladding. Coating SiC on the Zircaloy is preferred to a bulk design which has a brittle property, and Cr coating is preferred to adding more Cr to alloy since the addition of the material into the whole metallic structure induces sometimes detrimental effects for other properties such as the mechanical properties. In addition, coating requires less significant licensing efforts than a bulk design or composition change. However, coating may crack or fall off due to thermal expansion, thermal gradients or fretting, which will expose Zircaloy and going back to the original configurations. Regardless of these defects, remained parts of coating will reduce the oxidation during severe accidents in LWR. Furthermore, coating design will retain good properties of Zircaloy like mechanical property since it still keeps Zircaloy as a bulk material.

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